

★ WEEK 6 · PART 3 of 3

Shocks and Structure

Impulse Responses and Variance Decomposition

Correlated shocks · orthogonalization · IRF · FEVD · the bridge to Week 7

A VAR's real power is dynamic analysis: tracing how a single shock ripples through the whole system, and accounting for where each variable's uncertainty comes from.

3

PART 3 · SHOCKS & STRUCTURE

Tracing Shocks Through the System

Why shocks must be orthogonalized before we interpret them, how impulse responses trace their ripples, and how variance decomposition attributes uncertainty to its sources.

From Prediction to Propagation

The last, and most powerful, use of a VAR

1

Part 1 built the model, Part 2 tested links

We can fit a VAR, forecast jointly, and ask who predicts whom. Now we ask a deeper question: how does a shock move through the whole system over time?

2

But the shocks are tangled

A VAR's residuals are correlated — you cannot move one without the others. We must first untangle them into independent 'structural' shocks.

3

Then two tools finish the job

Impulse responses trace a shock's ripples; variance decomposition attributes each variable's uncertainty to its sources. Together they reveal the system's dynamics.

Part 3 in one line: untangle the shocks, then trace them (IRF) and budget them (FEVD).

A Subtlety: Correlated Shocks

Why we must 'orthogonalize' before interpreting

A VAR's residuals move together — a shock to one equation typically comes with a shock to the other. So 'shock only variable 1' is not something the raw model can even represent.

What goes wrong

- The residual covariance is not diagonal — shocks are correlated.
- Shocking Y1 alone ignores its usual companion move in Y2.
- Any 'response to a Y1 shock' is then ambiguous.

What we need

- A set of shocks that are mutually uncorrelated.
- Then each can be moved on its own, cleanly.
- That transformation is orthogonalization.

Why You Can't Just Shock One Variable

Correlation ties the shocks together

The raw residuals

- In real systems, a shock to one series rarely arrives alone.
- Interest-rate and equity shocks, say, tend to co-occur.
- The residual covariance captures exactly this co-movement.

The consequence

- An 'impulse' to one raw residual implies a partner impulse.
- So the naive response mixes two shocks together.
- We need independent shocks to get a clean answer.

Orthogonalization

Turn correlated shocks into independent ones

$$\boldsymbol{\varepsilon}_t = P \mathbf{u}_t, \quad \text{Cov}(\mathbf{u}_t) = I \quad \Rightarrow \quad \text{shocks } \mathbf{u}_t \text{ are uncorrelated}$$

The idea

- Write the correlated residuals as a mix of independent shocks.
- The structural shocks $\mathbf{u}_t$ are uncorrelated, with unit variance.
- Now each $\mathbf{u}$ can be moved on its own — a clean impulse.

Why it works

- Independence means shocking one leaves the others alone.
- The matrix P encodes how they combine into observed moves.
- The standard choice for P is the Cholesky factor.

The Cholesky Decomposition

A triangular square root of the covariance

$$\Sigma = P P^{\top} \quad P \text{ lower-triangular} \quad (\text{Cholesky decomposition})$$

What it gives

- A lower-triangular P with $\Sigma = P \cdot P^{\top}$.
- It maps the independent shocks u to the observed residuals.
- Triangular means an implicit ORDERING of the variables.

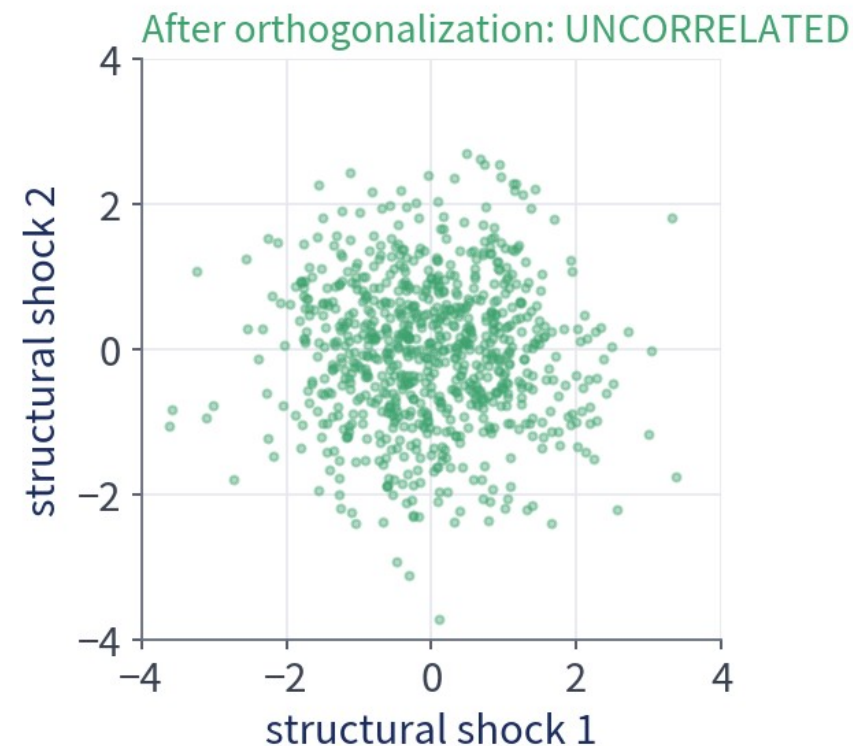
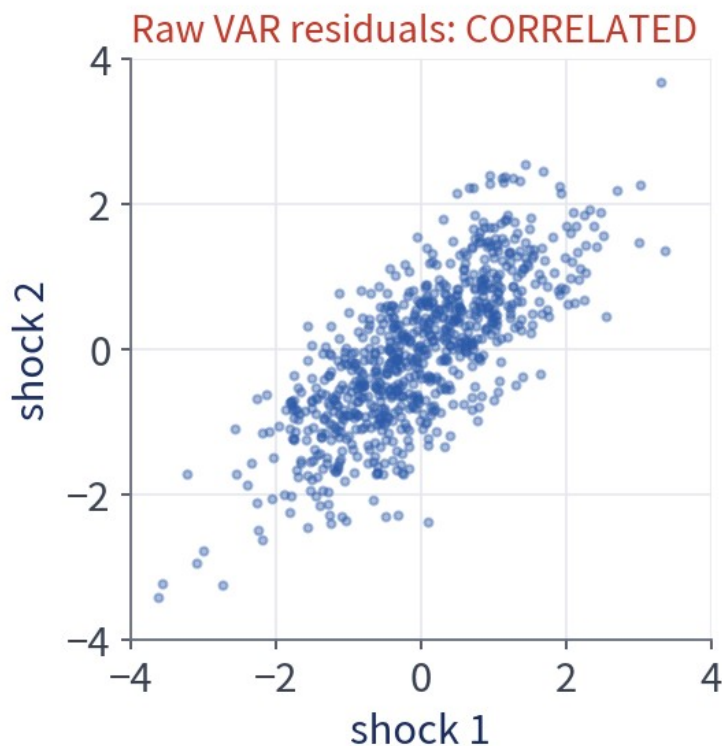
The catch

- The first variable gets all its own shock immediately.
- Later variables can respond within the period, but not vice versa.
- So the order you list variables in matters — a lot.

Untangling the Shocks

Raw VAR residuals form a tilted, correlated cloud (left). Orthogonalization rotates them into independent structural shocks (right) that can be moved one at a time.

Orthogonalization untangles correlated shocks into independent ones we can shock separately



The Ordering Problem

A choice that changes the answer

Cholesky orthogonalization assumes a causal ordering: the first variable can affect all others contemporaneously, but not the reverse. Different orderings give different impulse responses.

Why it matters

- The order encodes which shocks are allowed to hit first.
- Swap the order and the short-run responses can change.
- It is an assumption, not something the data decides.

How to handle it

- Order variables by economic reasoning (most exogenous first).
- Report sensitivity to the ordering, not a single result.
- For robust work, consider structural identification schemes.

Impulse Response Functions

How a one-time shock ripples through the system

An impulse response traces the effect of a single, isolated shock to one variable on every variable, over time. It is the clearest picture we have of a system's dynamics.

What it does

- Hit the system with one unit of a structural shock.
- Then watch every variable respond over the following periods.
- The path of each response is its impulse response function.

Why it is powerful

- It turns a matrix of coefficients into an intuitive story.
- You see the sign, the size, and how long effects last.
- It is the standard way to report VAR results.

The IRF, Defined

The response of i to shock j at horizon h

$$\text{IRF}_{ij}(h) = \frac{\partial y_{i,t+h}}{\partial u_{j,t}} = [A^h P]_{ij} \quad (\text{response of } i \text{ to shock } j \text{ at horizon } h)$$

Reading it

- Fix a shock to variable j at time t .
- The IRF is how variable i moves h periods later.
- It is computed from powers of A , scaled by the P matrix.

The shape it takes

- For a stable VAR, responses eventually decay to zero.
- They may overshoot, oscillate, or fade smoothly.
- The decay rate reflects the system's persistence.

Reading an IRF Plot

A grid of little stories

What each panel shows

- Each panel: response of one variable to one shock.
- The horizontal axis is the horizon after the shock.
- The vertical axis is the size of the response.

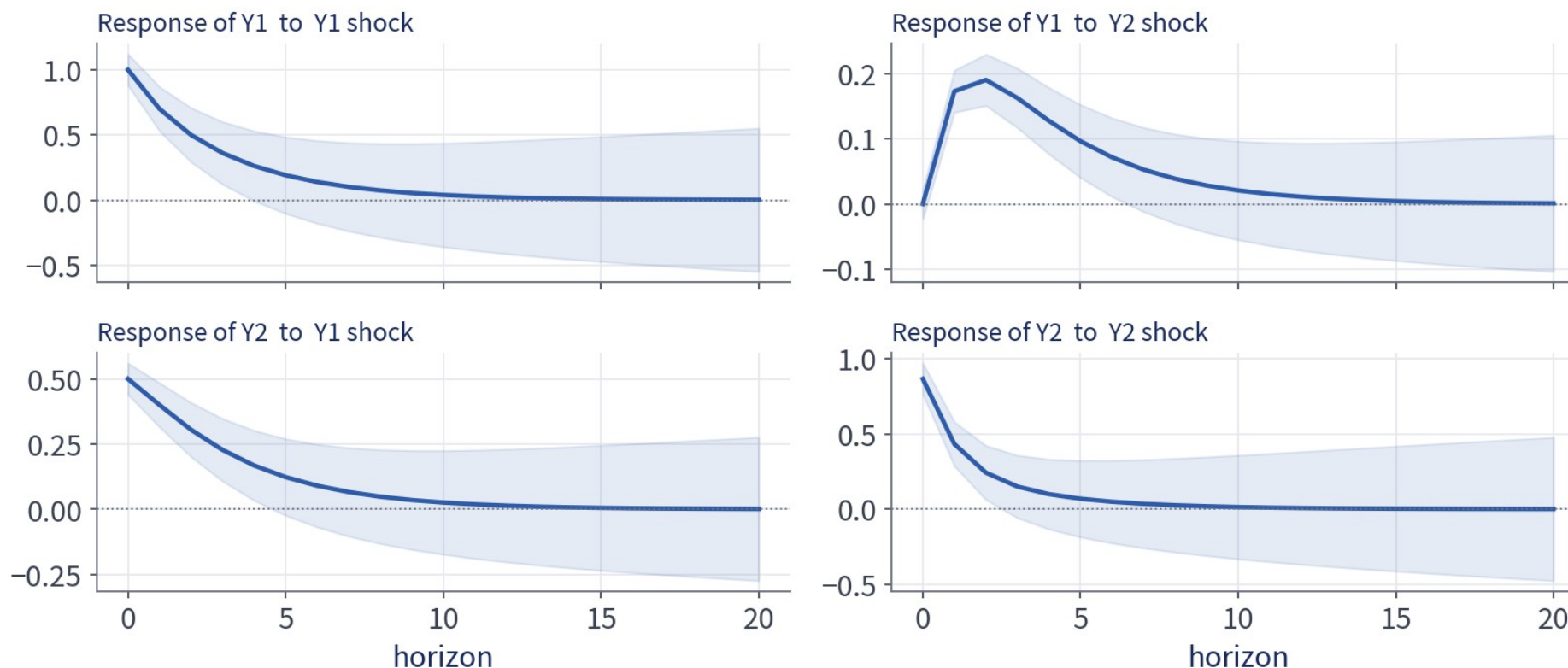
What to look for

- Sign: does the shock push the variable up or down?
- Persistence: how many periods until it fades?
- Bands: is the response statistically distinguishable from zero?

Tracing a Shock Through the System

Each panel traces one variable's response to one shock over the horizon, with uncertainty bands. Cross-panel responses show how the two series pass shocks to each other.

Impulse responses: how a one-time shock to each variable ripples through the system



What IRFs Reveal — and Their Bands

Reading the dynamics responsibly

The dynamics

- How strongly one series reacts to another's shock.
- Whether effects build, peak, and then decay.
- How long a disturbance keeps mattering.

The uncertainty

- IRFs are estimates — always plot confidence bands.
- A response is meaningful only if its band excludes zero.
- Bands usually widen at longer horizons.

Forecast Error Variance Decomposition

Where does a variable's uncertainty come from?

FEVD asks a budgeting question: of all the uncertainty in forecasting a variable, what fraction traces back to each shock? It turns the system into a variance budget by source.

The question

- Forecast a variable h steps ahead; it has some error variance.
- How much of that variance is due to its own shocks?
- How much to shocks from the other variables?

Why it complements IRF

- IRF shows the SHAPE of a shock's effect over time.
- FEVD shows the SHARE of variance each shock explains.
- Together: how it moves, and how much it matters.

The FEVD, Defined

A share of forecast-error variance

$$\text{FEVD}_{ij}(h) = \frac{\text{variance of } y_i \text{'s } h\text{-step error from shock } j}{\text{total } h\text{-step forecast error variance of } y_i}$$

Reading it

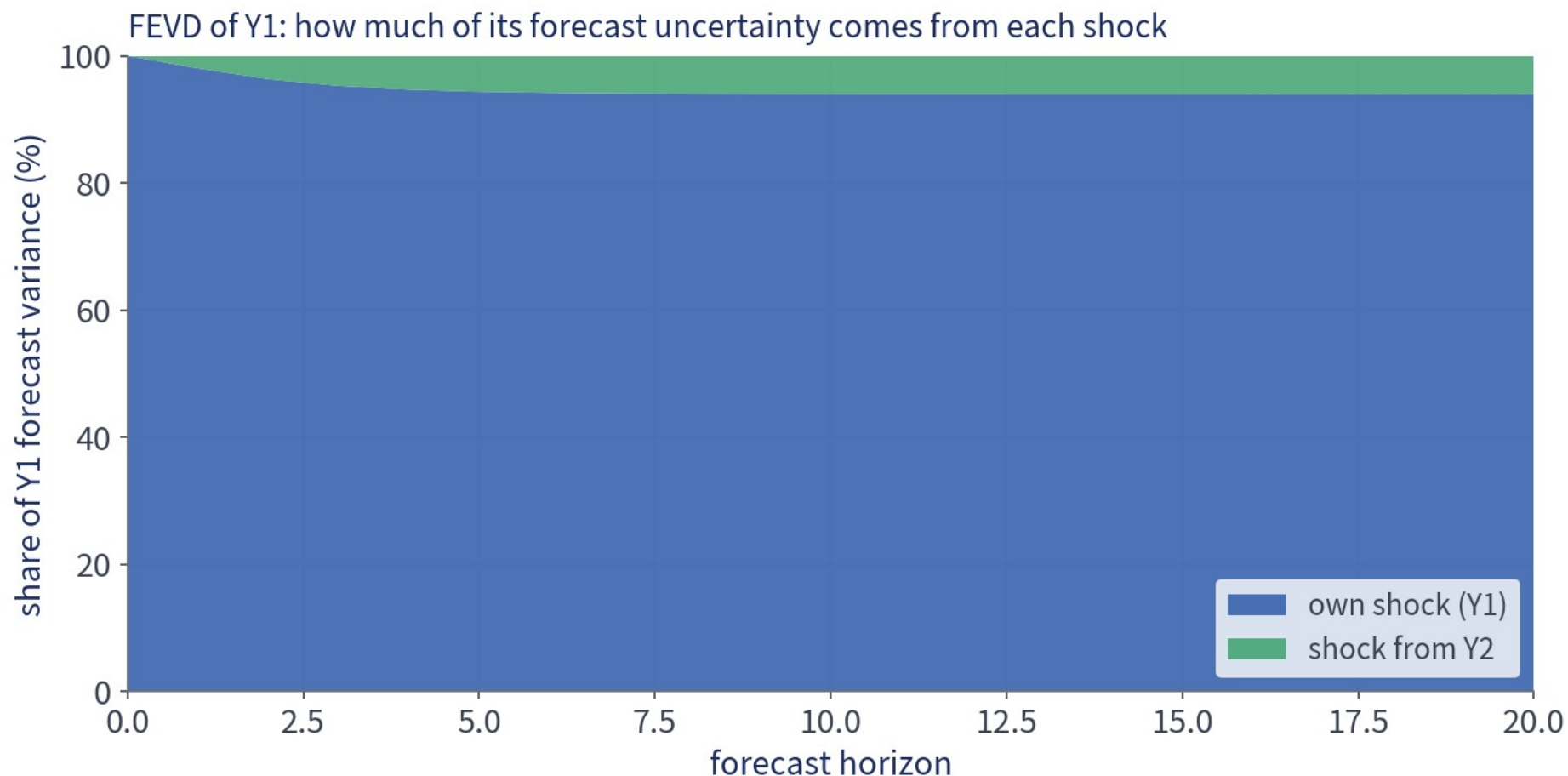
- The numerator: variance from shock j alone.
- The denominator: the variable's total forecast error variance.
- The ratio is a percentage, at each horizon h .

Its behavior

- At short horizons, a variable's own shock usually dominates.
- At longer horizons, other shocks' shares often grow.
- The shares always add up to 100% across shocks.

Where Forecast Uncertainty Comes From

For Y1, the share of forecast-error variance from its own shock versus the shock from Y2, across horizons. Early on, own shocks dominate; the other's share grows with horizon.



Reading a FEVD

A variance budget, horizon by horizon

What it tells you

- Which shocks drive a variable's uncertainty, and when.
- Whether a series is mostly self-driven or externally driven.
- How that balance shifts as the horizon lengthens.

How to use it

- Rank the sources of risk for each variable.
- Identify which series a shock elsewhere really matters for.
- Like IRF, it depends on the ordering — report sensitivity.

Granger, IRF, and FEVD

Three questions, one fitted VAR

Tool	Question it answers	What it outputs
Granger causality	Does X's past help predict Y?	yes/no, with a direction
Impulse response	How does a shock to X move Y over time?	a response path
Variance decomposition	What share of Y's uncertainty is due to X?	percentages, by horizon

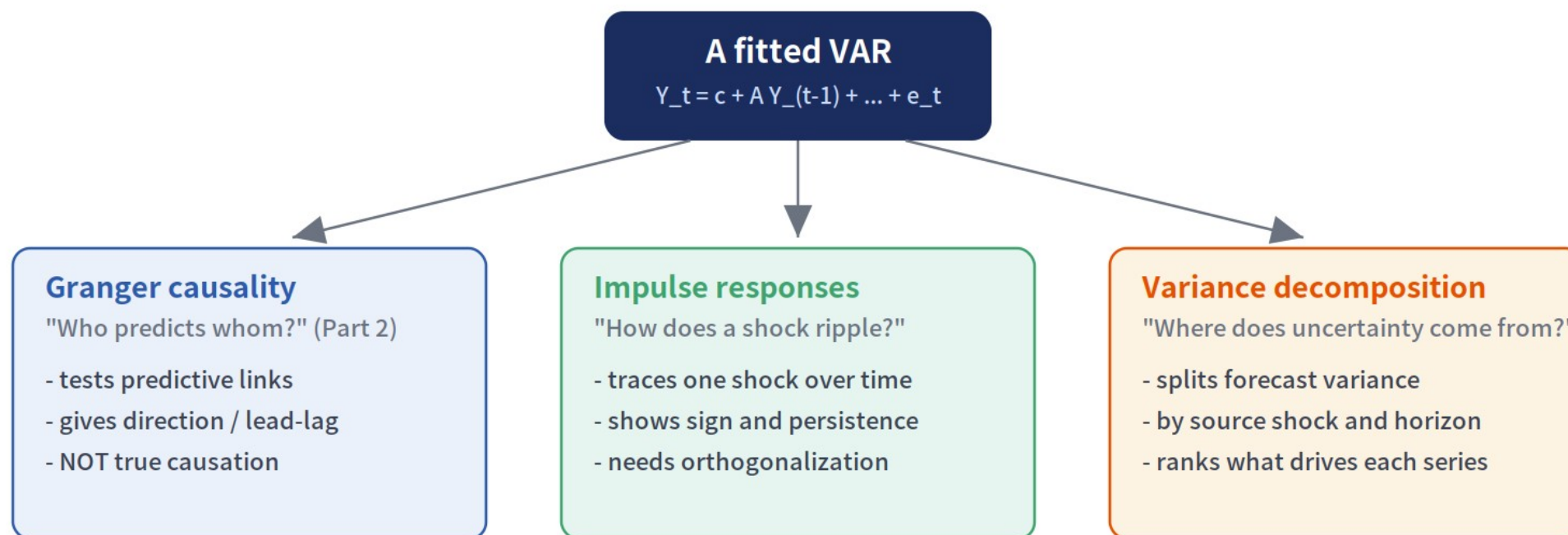
A test, a dynamic path, and a variance budget — three complementary views of the same system.

The VAR Analysis Toolkit

One fitted VAR answers three questions: who predicts whom (Granger), how shocks ripple (IRF), and where uncertainty originates (FEVD) — each a lens on the joint dynamics.

The VAR Analysis Toolkit

One fitted VAR answers three different questions - each a lens on the same joint dynamics.



The Complete VAR Picture

From data to forecasts to structure

What a VAR gives you

- Joint forecasts that respect cross-dynamics (Part 1).
- Directional predictive links via Granger tests (Part 2).
- Shock propagation and variance budgets via IRF and FEVD.

The honest caveats

- Reduced-form VARs describe, they do not prove causation.
- Structural claims (IRF, FEVD) hinge on the ordering.
- Keep the system small, stationary, and well-diagnosed.

From Co-Movement to a Tradable Anchor

How VAR sets up the next chapter

VAR models how series move together in the SHORT run. But some pairs share a LONG-run equilibrium they always return to — and that stable relationship is the anchor a pairs trade needs.

What VAR gave us

- A framework for joint dynamics and shock propagation.
- Tools to see which series lead, and how shocks spread.
- But standard VAR needs stationary (differenced) series.

Where Week 7 goes

- Cointegrated pairs share a long-run equilibrium in LEVELS.
- The error-correction model captures how they revert to it.
- That is the true statistical engine of pairs trading.

Impulse Responses and FEVD

Direct Claude Code to trace and budget shocks

GOAL From a fitted VAR, produce orthogonalized impulse responses and a variance decomposition, and probe the ordering.

[INPUT] Prompt · Claude Code

Write and run `irf_fevd_lab.py` that, from a fitted VAR on 2-3 series:

1. computes orthogonalized (Cholesky) impulse responses with confidence bands and plots the full IRF grid;
2. computes and stack-plots the FEVD for each variable across horizons;
3. RE-RUNS both with the variable order reversed and shows how much the results change.

Summarize what propagates to what, and how sensitive it is to the ordering.

How to work with it

- Read sign and persistence off each IRF panel.
- Check which shock dominates each FEVD.
- Note how reversing the order shifts short-run responses.
- This closes Week 6 and points to Week 7.

Deliverable: `irf_fevd_lab.py` + an IRF grid, a FEVD, and an ordering-sensitivity check.

Three Questions Before You Finish

If you can answer these, Week 6 has landed

Q1	Why must VAR shocks be orthogonalized before computing impulse responses?	Residuals are correlated
Q2	You reverse the variable order and the IRFs change. Why?	Cholesky ordering matters
Q3	A FEVD says Y2's shock explains 60% of Y1's variance. What does that mean?	Y2 drives Y1's uncertainty

All three come from correlated shocks, the ordering assumption, and the variance budget.

Three Pitfalls to Avoid

Where structural VAR analysis goes wrong

Common mistakes

- "I can shock one raw residual and read the response."
- "The impulse responses are the truth, whatever the order."
- "IRF and FEVD prove causation."

The reality

- Raw residuals are correlated — orthogonalize first.
- Cholesky results depend on the ordering — report sensitivity.
- They describe a reduced-form system, not proven mechanisms.

Week 6 in Three Ideas

Pulling the whole week together

1

A VAR models series jointly

Part 1: every series depends on the recent past of all series. The coefficient matrix carries the cross-dynamics, subject to stability and a parameter budget.

2

Granger tests prediction, not cause

Part 2: the F-test asks whether one series' past improves another's forecast — a directional predictive link, never a proof of causation.

3

IRF and FEVD reveal the structure

Part 3: orthogonalize the shocks, then trace their ripples (IRF) and budget the variance (FEVD) — the clearest view of the system's dynamics.

Next — Week 7: Cointegration and Error Correction, the long-run link that anchors pairs trading.